

Q1. Let three vectors $\vec{a} = \alpha\hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 5\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$ form a triangle such that $\vec{c} = \vec{a} - \vec{b}$ and the area of the triangle is $5\sqrt{6}$. If α is a positive real number, then $|\vec{c}|^2$ is equal to: 09 Apr 2024 (M)

- (1) 16 (2) 14
(3) 12 (4) 10

Q2. Let $\vec{OA} = 2\vec{a}$, $\vec{OB} = 6\vec{a} + 5\vec{b}$ and $\vec{OC} = 3\vec{b}$, where O is the origin. If the area of the parallelogram with adjacent sides $\vec{OA}$ and $\vec{OC}$ is 15 sq. units, then the area (in sq. units) of the quadrilateral $OABC$ is equal to : 09 Apr 2024 (M)

- (1) 32 (2) 40
(3) 38 (4) 35

Q3. Between the following two statements:

Statement I : Let $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$. Then the vector $\vec{r}$ satisfying $\vec{a} \times \vec{r} = \vec{a} \times \vec{b}$ and $\vec{a} \cdot \vec{r} = 0$ is of magnitude $\sqrt{10}$.

Statement II : In a triangle ABC , $\cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2}$. 09 Apr 2024 (E)

- (1) Statement I is incorrect but Statement II is correct.
(2) Both Statement I and Statement II are correct.
(3) Statement I is correct but Statement II is incorrect.
(4) Both Statement I and Statement II are incorrect.

Q4. Let $\vec{a} = 2\hat{i} + \alpha\hat{j} + \hat{k}$, $\vec{b} = -\hat{i} + \hat{k}$, $\vec{c} = \beta\hat{j} - \hat{k}$, where α and β are integers and $\alpha\beta = -6$. Let the values of the ordered pair (α, β) , for which the area of the parallelogram of diagonals $\vec{a} + \vec{b}$ and $\vec{b} + \vec{c}$ is $\frac{\sqrt{21}}{2}$, be (α_1, β_1) and (α_2, β_2) . Then $\alpha_1^2 + \beta_1^2 - \alpha_2\beta_2$ is equal to 09 Apr 2024 (E)

- (1) 19 (2) 17
(3) 24 (4) 21

Q5. Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} + 3\hat{j} - 5\hat{k}$ and $\vec{c} = 3\hat{i} - \hat{j} + \lambda\hat{k}$ be three vectors. Let $\vec{r}$ be a unit vector along $\vec{b} + \vec{c}$. If $\vec{r} \cdot \vec{a} = 3$, then 3λ is equal to: 08 Apr 2024 (E)

- (1) 21 (2) 30
(3) 25 (4) 27

Q6. The set of all α , for which the vectors $\vec{a} = \alpha\hat{i} + 6\hat{j} - 3\hat{k}$ and $\vec{b} = t\hat{i} - 2\hat{j} - 2\alpha t\hat{k}$ are inclined at an obtuse angle for all $t \in \mathbb{R}$, is 08 Apr 2024 (M)

- (1) $(-\frac{4}{3}, 1)$ (2) $[0, 1)$
(3) $(-\frac{4}{3}, 0]$ (4) $(-2, 0]$

Q7. Let $\vec{a} = 4\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = 11\hat{i} - \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $(\vec{a} + \vec{b}) \times \vec{c} = \vec{c} \times (-2\vec{a} + 3\vec{b})$. If $(2\vec{a} + 3\vec{b}) \cdot \vec{c} = 1670$, then $|\vec{c}|^2$ is equal to : 08 Apr 2024 (E)

- (1) 1609 (2) 1618
(3) 1600 (4) 1627

Q8. Let $\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}$, $\vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$ and $\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = (\vec{b} + \vec{c}) \times \vec{a}$ and $\vec{r} \cdot (\vec{b} - \vec{c}) = 0$, then $\frac{|593\vec{r} + 67\vec{a}|^2}{(593)^2}$ is equal to 08 Apr 2024 (M)

Q9. Let $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$, $\vec{b} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ and a vector $\vec{c}$ be such that $\vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times \vec{c} = \hat{i} + 8\hat{j} + 13\hat{k}$. If $\vec{a} \cdot \vec{c} = 13$, then $(24 - \vec{b} \cdot \vec{c})$ is equal to _____ **06 Apr 2024 (M)**

Q10. Let $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{b} = ((\vec{a} \times (\hat{i} + \hat{j})) \times \hat{i}) \times \hat{i}$. Then the square of the projection of $\vec{a}$ on $\vec{b}$ is : **06 Apr 2024 (E)**

- (1) $\frac{1}{3}$ (2) $\frac{2}{3}$
(3) 2 (4) $\frac{1}{5}$

Q11*. Let $\vec{a} = 6\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $|\vec{c}| \geq 6$, $\vec{a} \cdot \vec{c} = 6|\vec{c}|$, $|\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $\vec{a} \times \vec{b}$ and $\vec{c}$ is 60° , then $|(\vec{a} \times \vec{b}) \times \vec{c}|$ is equal to: **06 Apr 2024 (E)**

- (1) $\frac{9}{2}(6 - \sqrt{6})$ (2) $\frac{3}{2}\sqrt{6}$
(3) $\frac{9}{2}(6 + \sqrt{6})$ (4) $\frac{3}{2}\sqrt{3}$

Q12. Consider three vectors $\vec{a}, \vec{b}, \vec{c}$. Let $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $\vec{a} = \vec{b} \times \vec{c}$. If $\alpha \in [0, \frac{\pi}{3}]$ is the angle between the vectors $\vec{b}$ and $\vec{c}$, then the minimum value of $27|\vec{c} - \vec{a}|^2$ is equal to: **05 Apr 2024 (E)**

- (1) 110 (2) 124
(3) 121 (4) 105

Q13. Let $\vec{a} = \hat{i} - 3\hat{j} + 7\hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $(\vec{a} + 2\vec{b}) \times \vec{c} = 3(\vec{c} \times \vec{a})$. If $\vec{a} \cdot \vec{c} = 130$, then $\vec{b} \cdot \vec{c}$ is equal to _____ **05 Apr 2024 (M)**

Q14. Let $\vec{a} = 2\hat{i} + 5\hat{j} - \hat{k}$, $\vec{b} = 2\hat{i} - 2\hat{j} + 2\hat{k}$ and $\vec{c}$ be three vectors such that $(\vec{c} + \hat{i}) \times (\vec{a} + \vec{b} + \hat{i}) = \vec{a} \times (\vec{c} + \hat{i})$. If $\vec{a} \cdot \vec{c} = -29$, then $\vec{c} \cdot (-2\hat{i} + \hat{j} + \hat{k})$ is equal to: **05 Apr 2024 (E)**

- (1) 15 (2) 12
(3) 10 (4) 5

Q15. Let a unit vector which makes an angle of 60° with $2\hat{i} + 2\hat{j} - \hat{k}$ and angle 45° with $\hat{i} - \hat{k}$ be $\vec{C}$. Then $\vec{C} + \left(-\frac{1}{2}\hat{i} + \frac{1}{3\sqrt{2}}\hat{j} - \frac{\sqrt{2}}{3}\hat{k}\right)$ is : **04 Apr 2024 (M)**

- (1) $\frac{\sqrt{2}}{3}\hat{i} - \frac{1}{2}\hat{k}$ (2) $\left(\frac{1}{\sqrt{3}} + \frac{1}{2}\right)\hat{i} + \left(\frac{1}{\sqrt{3}} - \frac{1}{3\sqrt{2}}\right)\hat{j} + \left(\frac{1}{\sqrt{3}} + \frac{\sqrt{2}}{3}\right)\hat{k}$
(3) $\frac{\sqrt{2}}{3}\hat{i} + \frac{1}{3\sqrt{2}}\hat{j} - \frac{1}{2}\hat{k}$ (4) $-\frac{\sqrt{2}}{3}\hat{i} + \frac{\sqrt{2}}{3}\hat{j} + \left(\frac{1}{2} + \frac{2\sqrt{2}}{3}\right)\hat{k}$

Q16. Let ABC be a triangle of area $15\sqrt{2}$ and the vectors $\vec{AB} = \hat{i} + 2\hat{j} - 7\hat{k}$, $\vec{BC} = a\hat{i} + b\hat{j} + c\hat{k}$ and $\vec{AC} = 6\hat{i} + d\hat{j} - 2\hat{k}$, $d > 0$. Then the square of the length of the largest side of the triangle ABC is _____ **04 Apr 2024 (M)**

Q17. For $\lambda > 0$, let θ be the angle between the vectors $\vec{a} = \hat{i} + \lambda\hat{j} - 3\hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$. If the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are mutually perpendicular, then the value of $(14 \cos \theta)^2$ is equal to **04 Apr 2024 (E)**

- (1) 50 (2) 40
(3) 25 (4) 20

Q18*. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{c} = x\hat{i} + 2\hat{j} + 3\hat{k}$, $x \in \mathbb{R}$. If $\vec{d}$ is the unit vector in the direction of $\vec{b} + \vec{c}$ such that $\vec{a} \cdot \vec{d} = 1$, then $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is equal to **04 Apr 2024 (E)**

(1) 11

(3) 9

(2) 3

(4) 6

Q19. Let $\vec{a} = -5\hat{i} + \hat{j} - 3\hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - 4\hat{k}$ and $\vec{c} = \left(\left(\left(\vec{a} \times \vec{b} \right) \times \hat{i} \right) \times \hat{i} \right) \times \hat{i}$. Then $\vec{c} \cdot (-\hat{i} + \hat{j} + \hat{k})$ is equal

to

(1) -12

(3) -13

(2) -10

(4) -15

01 Feb 2024 (M)

Q20. Consider a $\triangle ABC$ where $A(1, 3, 2)$, $B(-2, 8, 0)$ and $C(3, 6, 7)$. If the angle bisector of $\angle BAC$ meets the line BC at D , then the length of the projection of the vector $\vec{AD}$ on the vector $\vec{AC}$ is:

(1) $\frac{37}{2\sqrt{38}}$ (3) $\frac{39}{2\sqrt{38}}$ (2) $\frac{\sqrt{38}}{2}$ (4) $\sqrt{19}$

01 Feb 2024 (E)

Q21. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = -\hat{i} - 8\hat{j} + 2\hat{k}$ and $\vec{c} = 4\hat{i} + c_2\hat{j} + c_3\hat{k}$ be three vectors such that $\vec{b} \times \vec{a} = \vec{c} \times \vec{a}$. If the angle between the vector $\vec{c}$ and the vector $3\hat{i} + 4\hat{j} + \hat{k}$ is θ , then the greatest integer less than or equal to $\tan^2 \theta$ is:

01 Feb 2024 (E)

Q22. Let $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = 4\hat{i} + \hat{j} + 7\hat{k}$ and $\vec{c} = \hat{i} - 3\hat{j} + 4\hat{k}$ be three vectors. If a vector $\vec{p}$ satisfies $\vec{p} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{p} \cdot \vec{a} = 0$, then $\vec{p} \cdot (\hat{i} - \hat{j} - \hat{k})$ is equal to

31 Jan 2024 (M)

(1) 24

(3) 28

(2) 36

(4) 32

Q23. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 4$ and $\vec{a} \cdot \vec{b} = 2$. If $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$ and the angle between $\vec{b}$ and $\vec{c}$ is α , then $192 \sin^2 \alpha$ is equal to _____

31 Jan 2024 (M)

Q24. Let $\vec{a} = 3\hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{c}$ be a vector such that $(\vec{a} + \vec{b}) \times \vec{c} = 2(\vec{a} \times \vec{b}) + 24\hat{j} - 6\hat{k}$ and $(\vec{a} - \vec{b} + \hat{i}) \cdot \vec{c} = -3$. Then $|\vec{c}|^2$ is equal to _____.

31 Jan 2024 (E)

Q25. Let $A(2, 3, 5)$ and $C(-3, 4, -2)$ be opposite vertices of a parallelogram $ABCD$ if the diagonal

 $\vec{BD} = \hat{i} + 2\hat{j} + 3\hat{k}$ then the area of the parallelogram is equal to

30 Jan 2024 (M)

(1) $\frac{1}{2}\sqrt{410}$ (3) $\frac{1}{2}\sqrt{586}$ (2) $\frac{1}{2}\sqrt{474}$ (4) $\frac{1}{2}\sqrt{306}$

Q26. Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ be two vectors such that $|\vec{a}| = 1$, $\vec{a} \cdot \vec{b} = 2$ and $|\vec{b}| = 4$. If

 $\vec{c} = 2(\vec{a} \times \vec{b}) - 3\vec{b}$, then the angle between $\vec{b}$ and $\vec{c}$ is equal to :

30 Jan 2024 (M)

(1) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (3) $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ (2) $\cos^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ (4) $\cos^{-1}\left(\frac{2}{3}\right)$

Q27. Let $\vec{a} = \hat{i} + \alpha\hat{j} + \beta\hat{k}$, $\alpha, \beta \in R$. Let a vector $\vec{b}$ be such that the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$ and $|\vec{b}|^2 = 6$. If $\vec{a} \cdot \vec{b} = 3\sqrt{2}$, then the value of $(\alpha^2 + \beta^2) |\vec{a} \times \vec{b}|^2$ is equal to 30 Jan 2024 (E)

(1) 90

(2) 75

(3) 95

(4) 85

Q28. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{b}| = 1$ and $|\vec{b} \times \vec{a}| = 2$. Then $|\left(\vec{b} \times \vec{a}\right) - \vec{b}|^2$ is equal to 30 Jan 2024 (E)

(1) 3

(2) 5

(3) 1

(4) 4

Q29. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-zero vectors such that $\vec{b}$ and $\vec{c}$ are non-collinear if $\vec{a} + 5\vec{b}$ is collinear with $\vec{c}$, $\vec{b} + 6\vec{c}$ is collinear with $\vec{a}$ and $\vec{a} + \alpha\vec{b} + \beta\vec{c} = \vec{0}$, then $\alpha + \beta$ is equal to 29 Jan 2024 (M)

(1) 35

(2) 30

(3) -30

(4) -25

Q30. Let $\vec{OA} = \vec{a}$, $\vec{OB} = 12\vec{a} + 4\vec{b}$ and $\vec{OC} = \vec{b}$, where O is the origin. If S is the parallelogram with adjacent sides OA and OC, then $\frac{\text{area of the quadrilateral OABC}}{\text{area of S}}$ is equal to 29 Jan 2024 (E)

(1) 6

(2) 10

(3) 7

(4) 8

Q31. Let a unit vector $\hat{u} = x\hat{i} + y\hat{j} + z\hat{k}$ make angles $\frac{\pi}{2}$, $\frac{\pi}{3}$ and $\frac{2\pi}{3}$ with the vectors $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k}$, $\frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$ and $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$ respectively. If $\vec{v} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$, then $|\hat{u} - \vec{v}|^2$ is equal to 29 Jan 2024 (E)

(1) $\frac{11}{2}$

(2) $\frac{5}{2}$

(3) 9

(4) 7

Q32. If $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 3(\hat{i} - \hat{j} + \hat{k})$ and $\vec{c}$ be the vector such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$, then $\vec{a} \cdot \left(\left(\vec{c} \times \vec{b} \right) - \vec{b} - \vec{c} \right)$ is equal to 27 Jan 2024 (M)

(1) 32

(2) 24

(3) 20

(4) 36

Q33. The least positive integral value of α , for which the angle between the vectors $\alpha\hat{i} - 2\hat{j} + 2\hat{k}$ and $\alpha\hat{i} + 2\alpha\hat{j} - 2\hat{k}$ is acute, is 27 Jan 2024 (M)

Q34. Let ABCD be a quadrilateral. If E and F are the mid points of the diagonals AC and BD respectively and $\left(\vec{AB} - \vec{BC} \right) + \left(\vec{AD} - \vec{DC} \right) = k \vec{FE}$, then k is equal to 15 Apr 2023 (M)

(1) 4

(2) -2

(3) 2

(4) -4

Q35. Let S be the set of all (λ, μ) for which the vectors $\lambda\hat{i} - \hat{j} + \hat{k}$, $\hat{j} + 2\hat{j} + \mu\hat{k}$ and $3\hat{i} - 4\hat{j} + 5\hat{k}$, where $\lambda - \mu = 5$, are coplanar, then $\sum_{(\lambda, \mu) \in S} 80(\lambda^2 + \mu^2)$ is equal to 15 Apr 2023 (M)

(1) 2210

(3) 2290

(2) 2130

(4) 2370

Q36. Let $|\vec{a}| = 2$, $|\vec{b}| = 3$ and the angle between the vectors $\vec{a}$ and $\vec{b}$ be $\frac{\pi}{4}$. Then $\left| \left(\vec{a} + 2\vec{b} \right) \times \left(2\vec{a} - 3\vec{b} \right) \right|^2$ is equal

(1) 441

(3) 841

(2) 482

(4) 882

13 Apr 2023 (E)

Q37. Let for a triangle ABC

$$\vec{AB} = -2\hat{i} + \hat{j} + 3\hat{k}$$

$$\vec{CB} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$$

$$\vec{CA} = 4\hat{i} + 3\hat{j} + \delta\hat{k}$$

If $\delta > 0$ and the area of the triangle ABC is $5\sqrt{6}$ then $\vec{CB} \cdot \vec{CA}$ is equal to

(1) 60

(3) 108

(2) 54

(4) 120

13 Apr 2023 (E)

Q38. Let $\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} - 3\hat{j} + 3\hat{k}$. If $\vec{b}$ is a vector such that $\vec{a} = \vec{b} \times \vec{c}$ and $|\vec{b}|^2 = 50$, then

$72 - |\vec{b} + \vec{c}|^2$ is equal to _____.

13 Apr 2023 (M)

Q39. Let $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. If a vector $\vec{d}$ satisfies $\vec{d} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{d} \cdot \vec{a} = 24$,

then $|\vec{d}|^2$ is equal to

(1) 323

(3) 313

(2) 423

(4) 413

13 Apr 2023 (M)

Q40. Let a, b, c be three distinct real numbers, none equal to one. If the vectors $a\hat{i} + \hat{j} + \hat{k}$, $\hat{i} + b\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} + c\hat{k}$ are coplanar, then $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ is equal to

(1) 2

(3) -2

(2) -1

(4) 1

12 Apr 2023 (M)

Q41. Let $\lambda \in \mathbb{Z}$, $\vec{a} = \lambda\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$. Let $\vec{c}$ be a vector such that

$(\vec{a} + \vec{b} + \vec{c}) \times \vec{c} = \vec{0}$, $\vec{a} \cdot \vec{c} = -17$ and $\vec{b} \cdot \vec{c} = -20$. Then $|\vec{c} \times (\lambda\hat{i} + \hat{j} + \hat{k})|^2$ is equal to

(1) 46

(3) 62

(2) 53

(4) 49

12 Apr 2023 (M)

Q42. Let the plane $x + 3y - 2z + 6 = 0$ meet the co-ordinate axes at the points A, B, C . If the orthocenter of the triangle ABC is $(\alpha, \beta, \frac{6}{7})$, then $98(\alpha + \beta)^2$ is equal to _____.

12 Apr 2023 (M)

Q43. Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} + \hat{j} - \hat{k}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = 11, \vec{b} \cdot (\vec{a} \times \vec{c}) = 27$ and $\vec{b} \cdot \vec{c} = -\sqrt{3}|\vec{b}|$, then $|\vec{a} \times \vec{c}|^2$ is equal to _____.

11 Apr 2023 (E)

Q44. Let $\vec{a}$ be a non-zero vector parallel to the line of intersection of the two planes described by $\hat{i} + \hat{j}, \hat{i} + \hat{k}$ and $\hat{i} - \hat{j}, \hat{j} - \hat{k}$. If θ is the angle between the vector $\vec{a}$ and the vector $\vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$ and $\vec{a} \cdot \vec{b} = 6$, then the ordered pair $(\theta, |\vec{a} \times \vec{b}|)$ is equal to

11 Apr 2023 (M)

- (1) $(\frac{\pi}{3}, 3\sqrt{6})$ (2) $(\frac{\pi}{4}, 3\sqrt{6})$
(3) $(\frac{\pi}{3}, 6)$ (4) $(\frac{\pi}{4}, 6)$

Q45*. If four distinct points with position vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are coplanar, then $[\vec{a}\vec{b}\vec{c}]$ is equal to

11 Apr 2023 (E)

- (1) $[\vec{d} \vec{b} \vec{a}] + [\vec{a} \vec{c} \vec{d}] + [\vec{db} \vec{c}]$ (2) $[\vec{a} \vec{d} \vec{b}] + [\vec{d} \vec{c} \vec{a}] + [\vec{d} \vec{b} \vec{c}]$
(3) $[\vec{d} \vec{c} \vec{a}] + [\vec{b} \vec{d} \vec{a}] + [\vec{c} \vec{d} \vec{b}]$ (4) $[\vec{b} \vec{c} \vec{d}] + [\vec{d} \vec{a} \vec{c}] + [\vec{d} \vec{b} \vec{a}]$

Q46. For any vector $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, with $10|a_i| < 1, i = 1, 2, 3$, consider the following statements:

- (A) : $\max\{|a_1|, |a_2|, |a_3|\} \leq |\vec{a}|$
(B) : $|\vec{a}| \leq 3 \max\{|a_1|, |a_2|, |a_3|\}$

11 Apr 2023 (M)

- (1) Only (B) is true (2) Only (A) is true
(3) Both (A) and (B) are true (4) Neither (A) nor (B) is true

Q47. Let $\vec{a} = 2\hat{i} + 7\hat{j} - \hat{k}$, $\vec{b} = 3\hat{i} + 5\hat{k}$ and $\vec{c} = \hat{i} - \hat{j} + 2\hat{k}$. Let $\vec{d}$ be a vector which is perpendicular to both $\vec{a}$ and $\vec{b}$, and $\vec{c} \cdot \vec{d} = 12$. Then $(-\hat{i} + \hat{j} - \hat{k}) \cdot (\vec{c} \times \vec{d})$ is equal to

10 Apr 2023 (E)

- (1) 24 (2) 44
(3) 42 (4) 48

Q48. If the points P and Q are respectively the circumcenter and the orthocentre of a ΔABC , then

10 Apr 2023 (E)

$\vec{PA} + \vec{PB} + \vec{PC}$ is equal to _____

- (1) $2\vec{QP}$ (2) $2\vec{PQ}$
(3) $\vec{PQ}$ (4) $\vec{QP}$

Q49. Let O be the origin and the position vector of the point P be $-\hat{i} - 2\hat{j} + 3\hat{k}$. If the position vectors of the points A, B and C are $-2\hat{i} + \hat{j} - 3\hat{k}$, $2\hat{i} + 4\hat{j} - 2\hat{k}$ and $-4\hat{i} + 2\hat{j} - \hat{k}$ respectively, then the projection of the vector $\vec{OP}$ on a vector perpendicular to the vectors $\vec{AB}$ and $\vec{AC}$ is

10 Apr 2023 (M)

- (1) 3 (2) $\frac{8}{3}$
(3) $\frac{7}{3}$ (4) $\frac{10}{3}$

Q50. An arc PQ of a circle subtends a right angle at its centre O . The mid point of the arc PQ is R . If $\overrightarrow{OP} = \vec{u}$, $\overrightarrow{OR} = \vec{v}$ and $\overrightarrow{OQ} = \alpha\vec{u} + \beta\vec{v}$, then α, β^2 , are the roots of the equation 10 Apr 2023 (M)

(1) $x^2 + x - 2 = 0$

(2) $x^2 - x - 2 = 0$

(3) $3x^2 - 2x - 1 = 0$

(4) $3x^2 + 2x - 1 = 0$

Q51. If the points with position vectors $\alpha\hat{i} + 10\hat{j} + 13\hat{k}$, $6\hat{i} + 11\hat{j} + 11\hat{k}$, $\frac{9}{2}\hat{i} + \beta\hat{j} - 8\hat{k}$ are collinear, then $(19\alpha - 6\beta)^2$ is equal to 08 Apr 2023 (M)

(1) 36

(2) 25

(3) 49

(4) 16

Q52. The area of the quadrilateral $ABCD$ with vertices $A(2, 1, 1)$, $B(1, 2, 5)$, $C(-2, -3, 5)$ and $D(1, -6, -7)$ is equal to 08 Apr 2023 (E)

(1) 48

(2) $8\sqrt{38}$

(3) 54

(4) $9\sqrt{38}$

Q53. Let the vectors $\vec{u}_1 = \hat{i} + \hat{j} + \hat{k}$, $\vec{u}_2 = \hat{i} + b\hat{j} + \hat{k}$, and $\vec{u}_3 = c\hat{i} + \hat{j} + \hat{k}$ be coplanar. If the vectors $\vec{v}_1 = (a+b)\hat{i} + c\hat{j} + c\hat{k}$, $\vec{v}_2 = a\hat{i} + (b+c)\hat{j} + a\hat{k}$ and $\vec{v}_3 = b\hat{i} + b\hat{j} + (c+a)\hat{k}$ are also coplanar, then $6(a+b+c)$ is equal to 08 Apr 2023 (E)

(1) 0

(2) 4

(3) 12

(4) 6

Q54. Let $\vec{a} = 6\hat{i} + 9\hat{j} + 12\hat{k}$, $\vec{b} = \alpha\hat{i} + 11\hat{j} - 2\hat{k}$ and $\vec{c}$ be vectors such that $\vec{a} \times \vec{c} = \vec{a} \times \vec{b}$. If $\vec{a} \cdot \vec{c} = -12$, and $\vec{c} \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 5$ then $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k})$ is equal to 08 Apr 2023 (M)

Q55. The sum of all values of α , for which the points whose position vectors are $\hat{i} - 2\hat{j} + 3\hat{k}$, $2\hat{i} - 3\hat{j} + 4\hat{k}$, $(\alpha + 1)\hat{i} + 2\hat{k}$ and $9\hat{i} + (\alpha - 8)\hat{j} + 6\hat{k}$ are coplanar, is equal to 06 Apr 2023 (E)

(1) -2

(2) 2

(3) 6

(4) 4

Q56. Let the position vectors of the points A, B, C and D be $5\hat{i} + 5\hat{j} + 2\lambda\hat{k}$, $\hat{i} + 2\hat{j} + 3\hat{k}$, $-2\hat{i} + \lambda\hat{j} + 4\hat{k}$ and $-\hat{i} + 5\hat{j} + 6\hat{k}$. Let the set $S = \{\lambda \in \mathbb{R} : \text{the points } A, B, C \text{ and } D \text{ are coplanar}\}$. The $\sum_{\lambda \in S} (\lambda + 2)^2$ is equal to 06 Apr 2023 (M)

(1) 25

(2) $\frac{37}{2}$

(3) 14

(4) 41

Q57. Let $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{b} = \hat{i} - 2\hat{j} - 2\hat{k}$ and $\vec{c} = -\hat{i} + 4\hat{j} + 3\hat{k}$. If $\vec{d}$ is a vector perpendicular to both $\vec{b}$ and $\vec{c}$, and $\vec{a} \cdot \vec{d} = 18$, then $|\vec{a} \times \vec{d}|^2$ is equal to 06 Apr 2023 (M)

(1) 640

(2) 680

(3) 720

(4) 760

Q58* Let the vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ represent three coterminal edges of a parallelepiped of volume V . Then the volume of the parallelepiped, whose coterminal edges are represented by $\vec{a}$, $\vec{b} + \vec{c}$ and $\vec{a} + 2\vec{b} + 3\vec{c}$ is equal to

06 Apr 2023 (E)

- (1) $2V$
(3) V

- (2) $6V$
(4) $3V$

Q59. Let $\vec{a} = 5\hat{i} - \hat{j} - 3\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} + 5\hat{k}$ be two vectors. Then which one of the following statements is TRUE?

01 Feb 2023 (E)

- (1) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{-13}{\sqrt{35}}$ and the direction of the projection vector is opposite to the direction of $\vec{b}$
(2) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{-17}{\sqrt{35}}$ and the direction of the projection vector is opposite to the direction of $\vec{b}$
(3) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{17}{\sqrt{35}}$ and the direction of the projection vector is opposite to the direction of $\vec{b}$
(4) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{13}{\sqrt{35}}$ and the direction of the projection vector is opposite to the direction of $\vec{a}$

Q60. $A(2, 6, 2)$, $B(-4, 0, \lambda)$, $C(2, 3, -1)$ and $D(4, 5, 0)$, $|\lambda| \leq 5$ are the vertices of a quadrilateral $ABCD$. If its area is 18 square units, then $5 - 6\lambda$ is equal to _____.

01 Feb 2023 (M)

Q61. Let $\vec{a} = 2\hat{i} - 7\hat{j} + 5\hat{k}$, $\vec{b} = \hat{i} + \hat{k}$ and $\vec{c} = \hat{i} + 2\hat{j} - 3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = \vec{c} \times \vec{a}$ and $\vec{r} \cdot \vec{b} = 0$, then $|\vec{r}|$ is equal to:

01 Feb 2023 (E)

- (1) $\frac{11}{7}\sqrt{2}$
(3) $\frac{11}{5}\sqrt{2}$

- (2) $\frac{11}{7}$
(4) $\frac{\sqrt{914}}{7}$

Q62* Let $\vec{v} = \alpha\hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{w} = 2\alpha\hat{i} + \hat{j} - \hat{k}$, and $\vec{u}$ be a vector such that $|\vec{u}| = \alpha > 0$. If the minimum value of the scalar triple product $[\vec{u} \vec{v} \vec{w}]$ is $-\alpha\sqrt{3401}$, and $|\vec{u} \cdot \hat{i}|^2 = \frac{m}{n}$ where m and n are coprime natural numbers, then $m + n$ is equal to _____.

01 Feb 2023 (M)

Q63. Let $\vec{a}$ and $\vec{b}$ be two vector such that $|\vec{a}| = \sqrt{14}$, $|\vec{b}| = \sqrt{6}$ and $|\vec{a} \times \vec{b}| = \sqrt{48}$. Then $(\vec{a} \cdot \vec{b})^2$ is equal to _____.

31 Jan 2023 (M)

Q64. Let $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$, and $\vec{b}$ and $\vec{c}$ be two nonzero vectors such that $|\vec{a} + \vec{b} + \vec{c}| = |\vec{a} + \vec{b} - \vec{c}|$ and $\vec{b} \cdot \vec{c} = 0$.

Consider the following two statement:

- (A) $|\vec{a} + \lambda\vec{c}| \geq |\vec{a}|$ for all $\lambda \in \mathbb{R}$.
(B) $\vec{a}$ and $\vec{c}$ are always parallel

31 Jan 2023 (M)

- (1) only (B) is correct
(3) only (A) is correct

- (2) neither (A) nor (B) is correct
(4) both (A) and (B) are correct.

Q65. Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = \hat{i} - \hat{j} + 2\hat{k}$ and $\vec{c} = 5\hat{i} - 3\hat{j} + 3\hat{k}$, be there(three) vector. If $\vec{r}$ is a vector such that, $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$, then $25|\vec{r}|^2$ is equal to

31 Jan 2023 (E)

(1) 560

(3) 449

(2) 339

(4) 336

Q66. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $|\vec{a}| = \sqrt{31}, 4|\vec{b}| = |\vec{c}| = 2$ and $2(\vec{a} \times \vec{b}) = 3(\vec{c} \times \vec{a})$. If the angle between $\vec{b}$ and $\vec{c}$ is $\frac{2\pi}{3}$, then $\left(\frac{\vec{a} \times \vec{c}}{\vec{a} \cdot \vec{b}}\right)^2$ is equal to _____. 31 Jan 2023 (E)

Q67*. Let $\vec{a}$ and $\vec{b}$ be two vectors. Let $|\vec{a}| = 1, |\vec{b}| = 4$ and $\vec{a} \cdot \vec{b} = 2$. If $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$, then the value of $\vec{b} \cdot \vec{c}$ is 30 Jan 2023 (E)

(1) -24

(2) -48

(3) -84

(4) -60

Q68*. If $\vec{a}, \vec{b}, \vec{c}$ are three non-zero vectors and $\hat{n}$ is a unit vector perpendicular to $\vec{c}$ such that $\vec{a} = \alpha\vec{b} - \hat{n}, (\alpha \neq 0)$ and $\vec{b} \cdot \vec{c} = 12$, then $|\vec{c} \times (\vec{a} \times \vec{b})|$ is equal to: 30 Jan 2023 (M)

(1) 15

(2) 9

(3) 12

(4) 6

Q69*. Let $\lambda \in \mathbb{R}, \vec{a} = \lambda\hat{i} + 2\hat{j} - 3\hat{k}, \vec{b} = \hat{i} - \lambda\hat{j} + 2\hat{k}$. If $\left((\vec{a} + \vec{b}) \times (\vec{a} \times \vec{b})\right) \times (\vec{a} - \vec{b}) = 8\hat{i} - 40\hat{j} - 24\hat{k}$ then $\left|\lambda(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})\right|^2$ is equal to 30 Jan 2023 (E)

(1) 140

(2) 132

(3) 144

(4) 136

Q70. If the vectors $\vec{a} = \lambda\hat{i} + \mu\hat{j} + 4\hat{k}, \vec{b} = -2\hat{i} + 4\hat{j} - 2\hat{k}$ and $\vec{c} = 2\hat{i} + 3\hat{j} + \hat{k}$ are coplanar and the projection of $\vec{a}$ on the vector $\vec{b}$ is $\sqrt{54}$ units, then the sum of all possible values of $\lambda + \mu$ is equal to 29 Jan 2023 (M)

(1) 0

(2) 6

(3) 24

(4) 18

Q71. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-zero non-coplanar vectors. Let the position vectors of four points A, B, C and D be $\vec{a} - \vec{b} + \vec{c}, \lambda\vec{a} - 3\vec{b} + 4\vec{c}, -\vec{a} + 2\vec{b} - 3\vec{c}$ and $2\vec{a} - 4\vec{b} + 6\vec{c}$ respectively. If $\vec{AB}, \vec{AC}$ and $\vec{AD}$ are coplanar, then λ is: 29 Jan 2023 (M)

Q72. If $\vec{a} = \hat{i} + 2\hat{k}, \vec{b} = \hat{i} + \hat{j} + \hat{k}, \vec{c} = 7\hat{i} - 3\hat{j} + 4\hat{k}, \vec{r} \times \vec{b} + \vec{b} \times \vec{c} = \vec{0}$ and $\vec{r} \cdot \vec{a} = 0$ then $\vec{r} \cdot \vec{c}$ is equal to: 29 Jan 2023 (E)

(1) 34

(2) 12

(3) 36

(4) 30

Q73*. Let $\vec{a} = 4\hat{i} + 3\hat{j}$ and $\vec{b} = 3\hat{i} - 4\hat{j} + 5\hat{k}$ and $\vec{c}$ is a vector such that $\vec{c} \cdot (\vec{a} \times \vec{b}) + 25 = 0, \vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = 4$ and projection of $\vec{c}$ on $\vec{a}$ is 1, then the projection of $\vec{c}$ on $\vec{b}$ equals: 29 Jan 2023 (E)

(1) $\frac{5}{\sqrt{2}}$
(3) $\frac{1}{\sqrt{2}}$

(2) $\frac{1}{5}$
(4) $\frac{3}{\sqrt{2}}$

Q74. If the four points, whose position vectors are $3\hat{i} - 4\hat{j} + 2\hat{k}$, $\hat{i} + 2\hat{j} - \hat{k}$, $-2\hat{i} - \hat{j} + 3\hat{k}$ and $5\hat{i} - 2\alpha\hat{j} + 4\hat{k}$ are coplanar, then α is equal to

25 Jan 2023 (E)

(1) $\frac{73}{17}$
(3) $-\frac{73}{17}$

(2) $-\frac{107}{17}$
(4) $\frac{107}{17}$

Q75. The vector $\vec{a} = -\hat{i} + 2\hat{j} + \hat{k}$ is rotated through a right angle, passing through the y -axis in its way and the resulting vector is $\vec{b}$. Then the projection of $3\vec{a} + \sqrt{2}\vec{b}$ on $\vec{c} = 5\hat{i} + 4\hat{j} + 3\hat{k}$ is

25 Jan 2023 (M)

(1) $3\sqrt{2}$
(3) $\sqrt{6}$

(2) 1
(4) $2\sqrt{3}$

Q76. Let $\vec{a} = -\hat{i} - \hat{j} + \hat{k}$, $\vec{a} \cdot \vec{b} = 1$ and $\vec{a} \times \vec{b} = \hat{i} - \hat{j}$. Then $\vec{a} - 6\vec{b}$ is equal to

25 Jan 2023 (E)

(1) $3(\hat{i} - \hat{j} - \hat{k})$

(2) $3(\hat{i} + \hat{j} + \hat{k})$

(3) $3(\hat{i} - \hat{j} + \hat{k})$

(4) $3(\hat{i} + \hat{j} - \hat{k})$

Q77*. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non zero vectors such that $\vec{b} \cdot \vec{c} = 0$ and $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b} - \vec{c}}{2}$. If $\vec{d}$ be a vector such

that $\vec{b} \cdot \vec{d} = \vec{a} \cdot \vec{b}$, then $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d})$ is equal to

25 Jan 2023 (M)

(1) $\frac{3}{4}$
(3) $-\frac{1}{4}$

(2) $\frac{1}{2}$
(4) $\frac{1}{4}$

Q78. Let $\vec{\alpha} = 4\hat{i} + 3\hat{j} + 5\hat{k}$ and $\vec{\beta} = \hat{i} + 2\hat{j} - 4\hat{k}$. Let $\vec{\beta}_1$ be parallel to $\vec{\alpha}$ and $\vec{\beta}_2$ be perpendicular to $\vec{\alpha}$. If $\vec{\beta} = \vec{\beta}_1 + \vec{\beta}_2$, then the value of $5\vec{\beta}_2 \cdot (\hat{i} + \hat{j} + \hat{k})$ is

24 Jan 2023 (E)

(1) 6
(3) 7

(2) 11
(4) 9

Q79. Let PQR be a triangle. The points A , B and C are on the sides QR , RP and PQ respectively such that

$$\frac{QA}{AR} = \frac{RB}{BP} = \frac{PC}{CQ} = \frac{1}{2}. \text{ Then } \frac{\text{Area}(\Delta PQR)}{\text{Area}(\Delta ABC)} \text{ is equal to}$$

24 Jan 2023 (M)

(1) 4
(3) 2

(2) 1
(4) $\frac{5}{2}$

Q80. Let $\vec{a} = \hat{i} + 2\hat{j} + \lambda\hat{k}$, $\vec{b} = 3\hat{i} - 5\hat{j} - \lambda\hat{k}$, $\vec{a} \cdot \vec{c} = 7$, $2(\vec{b} \cdot \vec{c}) + 43 = 0$, $\vec{a} \times \vec{c} = \vec{b} \times \vec{c}$, then $\vec{a} \cdot \vec{b}$ is equal to

24 Jan 2023 (E)

Q81. Let $\vec{u} = \hat{i} - \hat{j} - 2\hat{k}$, $\vec{v} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{v} \cdot \vec{w} = 2$ and $\vec{v} \times \vec{w} = \vec{u} + \lambda\vec{v}$, then $\vec{u} \cdot \vec{w}$ is equal to

24 Jan 2023 (M)

(1) 1
(3) 2

(2) $\frac{3}{2}$
(4) $-\frac{2}{3}$

Q82. If $(2, 3, 9)$, $(5, 2, 1)$, $(1, \lambda, 8)$ and $(\lambda, 2, 3)$ are coplanar, then the product of all possible values of λ is

29 Jul 2022 (E)

(1) $\frac{21}{2}$
(3) $\frac{57}{8}$

(2) $\frac{59}{8}$
(4) $\frac{95}{8}$

Q83. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + 2|\vec{b}|^2$, $\vec{a} \cdot \vec{b} = 3$ and $|\vec{a} \times \vec{b}|^2 = 75$. Then $|\vec{a}|^2$ is equal to _____.

29 Jul 2022 (E)

Q84*. Let $\vec{a} = 3\hat{i} + \hat{j}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. Let $\vec{c}$ be a vector satisfying $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \lambda\vec{c}$. If $\vec{b}$ and $\vec{c}$ are non-parallel, then the value of λ is

29 Jul 2022 (M)

(1) -5
(3) 1

(2) 5
(4) -1

Q85*. Let $\vec{a}, \vec{b}, \vec{c}$ be three coplanar concurrent vectors such that angles between any two of them is same. If the product of their magnitudes is 14 and $(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) + (\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) + (\vec{c} \times \vec{a}) \cdot (\vec{a} \times \vec{b}) = 168$ then $|\vec{a}| + |\vec{b}| + |\vec{c}|$ is equal to

29 Jul 2022 (E)

(1) 10
(3) 16

(2) 14
(4) 18

Q86. Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that the angle between them is $\frac{\pi}{4}$. If θ is the angle between the vectors $(\hat{a} + \hat{b})$ and $(\hat{a} + 2\hat{b} + 2(\hat{a} \times \hat{b}))$ then the value of $164 \cos^2 \theta$ is equal to

29 Jul 2022 (M)

(1) $90 + 27\sqrt{2}$
(3) $90 + 3\sqrt{2}$

(2) $45 + 18\sqrt{2}$
(4) $54 + 90\sqrt{2}$

Q87. Let the vectors $\vec{a} = (1+t)\hat{i} + (1-t)\hat{j} + \hat{k}$, $\vec{b} = (1-t)\hat{i} + (1+t)\hat{j} + 2\hat{k}$ and $\vec{c} = t\hat{i} - t\hat{j} + \hat{k}$, $t \in \mathbb{R}$ be such that for $\alpha, \beta, \gamma \in \mathbb{R}$, $\alpha\vec{a} + \beta\vec{b} + \gamma\vec{c} = \vec{0} \Rightarrow \alpha = \beta = \gamma = 0$. Then, the set of all values of t is

28 Jul 2022 (M)

(1) a non-empty finite set
(3) equal to $\mathbb{R} - \{0\}$

(2) equal to $\mathbb{N}$
(4) equal to $\mathbb{R}$

Q88. Let a vector $\vec{a}$ has a magnitude 9. Let a vector $\vec{b}$ be such that for every $(x, y) \in \mathbb{R} \times \mathbb{R} - \{(0, 0)\}$, the vector $(x\vec{a} + y\vec{b})$ is perpendicular to the vector $(6y\vec{a} - 18x\vec{b})$. Then the value of $|\vec{a} \times \vec{b}|$ is equal to

28 Jul 2022 (M)

(1) $9\sqrt{3}$
(3) 9

(2) $27\sqrt{3}$
(4) 81

Q89. Let S be the set of all $a \in \mathbb{R}$ for which the angle between the vectors $\vec{u} = a(\log_e b)\hat{i} - 6\hat{j} + 3\hat{k}$ and $\vec{v} = (\log_e b)\hat{i} + 2\hat{j} + 2a(\log_e b)\hat{k}$, $(b > 1)$ is acute. Then S is equal to

28 Jul 2022 (E)

(1) $(-\infty, -\frac{4}{3})$
(3) $(-\frac{4}{3}, 0)$

(2) $\emptyset$
(4) $(\frac{12}{7}, \infty)$

Q90. Let $\vec{a} = \alpha\hat{i} + \hat{j} + \beta\hat{k}$ and $\vec{b} = 3\hat{i} - 5\hat{j} + 4\hat{k}$ be two vectors, such that $\vec{a} \times \vec{b} = -\hat{i} + 9\hat{j} + 12\hat{k}$. Then the projection of $\vec{b} - 2\vec{a}$ on $\vec{b} + \vec{a}$ is equal to

27 Jul 2022 (M)

(1) 2

(3) 9

(2) $\frac{39}{5}$ (4) $\frac{46}{5}$

Q91* Let $\vec{a} = 2\hat{i} - \hat{j} + 5\hat{k}$ and $\vec{b} = \alpha\hat{i} + \beta\hat{j} + 2\hat{k}$. If $\left(\left(\vec{a} \times \vec{b}\right) \times \hat{i}\right) \cdot \hat{k} = \frac{23}{2}$, then $|\vec{b} \times 2\hat{j}|$ is equal to

27 Jul 2022 (M)

(1) 4

(2) 5

(3) $\sqrt{21}$ (4) $\sqrt{17}$

Q92* Let $\vec{a}, \vec{b}, \vec{c}$ be three non-coplanar vectors such that $\vec{a} \times \vec{b} = 4\vec{c}$, $\vec{b} \times \vec{c} = 9\vec{a}$ and $\vec{c} \times \vec{a} = \alpha\vec{b}$, $\alpha > 0$

27 Jul 2022 (E)

If $|\vec{a}| + |\vec{b}| + |\vec{c}| = 36$, then α is equal to _____.

Q93. Let $\vec{a} = \alpha\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - \alpha\hat{k}$, $\alpha > 0$. If the projection of $\vec{a} \times \vec{b}$ on the vector $-\hat{i} + 2\hat{j} - 2\hat{k}$ is 30, then α is equal to

26 Jul 2022 (M)

(1) $\frac{15}{2}$

(2) 8

(3) $\frac{13}{2}$

(4) 7

Q94. Let ABC be a triangle such that $\vec{BC} = \vec{a}$, $\vec{CA} = \vec{b}$, $\vec{AB} = \vec{c}$, $|\vec{a}| = 6\sqrt{2}$, $|\vec{b}| = 2\sqrt{3}$ and $\vec{b} \cdot \vec{c} = 12$. Consider the statements :

(S1): $\left|(\vec{a} \times \vec{b}) + (\vec{c} \times \vec{b})\right| - |\vec{c}| = 6(2\sqrt{2} - 1)$

(S2): $\angle ABC = \cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$. Then

25 Jul 2022 (M)

(1) both (S1) and (S2) are true

(2) only (S1) is true

(3) only (S2) is true

(4) both (S1) and (S2) are false

Q95. Let $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$ and let $\vec{b}$ be a vector such that $\vec{a} \times \vec{b} = 2\hat{i} - \hat{k}$ and $\vec{a} \cdot \vec{b} = 3$. Then the projection of $\vec{b}$ on the vector $\vec{a} - \vec{b}$ is:

25 Jul 2022 (E)

(1) $\frac{2}{\sqrt{21}}$ (2) $2\sqrt{\frac{3}{7}}$ (3) $\frac{2}{3}\sqrt{\frac{7}{3}}$ (4) $\frac{2}{3}$

Q96. Let $\vec{a} = \alpha\hat{i} + 3\hat{j} - \hat{k}$, $\vec{b} = 3\hat{i} - \beta\hat{j} + 4\hat{k}$ and $\vec{c} = \hat{i} + 2\hat{j} - 2\hat{k}$ where $\alpha, \beta \in \mathbb{R}$ be three vectors. If the projection of $\vec{a}$ on $\vec{c}$ is $\frac{10}{3}$ and $\vec{b} \times \vec{c} = -6\hat{i} + 10\hat{j} + 7\hat{k}$, then the value of $\alpha + \beta$ equal to

29 Jun 2022 (M)

(1) 3

(2) 4

(3) 5

(4) 6

Q97. Let $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} \times (\vec{b} + \vec{c}) = \vec{0}$, then the value of $3(\vec{c} \cdot \vec{a})$ is equal to _____.

29 Jun 2022 (E)

Q98. Let A, B, C be three points whose position vectors respectively are:

$$\vec{a} = \hat{i} + 4\hat{j} + 3\hat{k}$$

$$\vec{b} = 2\hat{i} + \alpha\hat{j} + 4\hat{k}, \alpha \in \mathbb{R}$$

$$\vec{c} = 3\hat{i} - 2\hat{j} + 5\hat{k}$$

If α is the smallest positive integer for which $\vec{a}, \vec{b}, \vec{c}$ are non-collinear, then the length of the median, $\triangle ABC$, through A is:

29 Jun 2022 (E)

(1) $\frac{\sqrt{82}}{2}$

(2) $\frac{\sqrt{62}}{2}$

(3) $\frac{\sqrt{69}}{2}$

(4) $\frac{\sqrt{66}}{2}$

Q99. Let $\vec{a}$ be a vector which is perpendicular to the vector $3\hat{i} + \frac{1}{2}\hat{j} + 2\hat{k}$. If $\vec{a} \times (2\hat{i} + \hat{k}) = 2\hat{i} - 13\hat{j} - 4\hat{k}$, then the projection of the vector $\vec{a}$ on the vector $2\hat{i} + 2\hat{j} + \hat{k}$ is

28 Jun 2022 (E)

(1) $\frac{1}{3}$

(2) 1

(3) $\frac{5}{3}$

(4) $\frac{7}{3}$

Q100. If $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}$, $\vec{b} = 3\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ are coplanar vectors and $\vec{a} \cdot \vec{c} = 5, \vec{b} \perp \vec{c}$, then $122(c_1 + c_2 + c_3)$ is equal to _____.

28 Jun 2022 (M)

Q101. Let $\vec{a} = \alpha\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{b} = -2\hat{i} + \alpha\hat{j} + \hat{k}$, where $\alpha \in \mathbb{R}$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $\sqrt{15(\alpha^2 + 4)}$, then the value of $2|\vec{a}|^2 + \left(\vec{a} \cdot \vec{b}\right)|\vec{b}|^2$ is equal to

28 Jun 2022 (E)

(1) 10

(2) 7

(3) 9

(4) 14

Q102. Let $\vec{a}$ and $\vec{b}$ be the vectors along the diagonal of a parallelogram having area $2\sqrt{2}$. Let the angle between $\vec{a}$ and $\vec{b}$ be acute. $|\vec{a}| = 1$ and $|\vec{a} \cdot \vec{b}| = |\vec{a} \times \vec{b}|$. If $\vec{c} = 2\sqrt{2}(\vec{a} \times \vec{b}) - 2\vec{b}$, then an angle between $\vec{b}$ and $\vec{c}$ is

27 Jun 2022 (E)

(1) $\frac{\pi}{4}$

(2) $\frac{5\pi}{6}$

(3) $\frac{\pi}{3}$

(4) $\frac{3\pi}{4}$

Q103. Let $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} - 3\hat{j} + 2\hat{k}$. Then the number of vectors $\vec{b}$ such that $\vec{b} \times \vec{c} = \vec{a}$ and $|\vec{b}| \in \{1, 2, \dots, 10\}$ is

27 Jun 2022 (M)

(1) 0

(2) 1

(3) 2

(4) 3

Q104. Let $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}, \vec{b} = 2\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} + \hat{k}$ be the three given vectors. Let $\vec{v}$ be a vector in the plane of $\vec{a}$ and $\vec{b}$ whose projection on $\vec{c}$ is $\frac{2}{\sqrt{3}}$. If $\vec{v} \cdot (\hat{i} + \hat{k}) = 7$, then $\vec{v} \cdot (\hat{i} + \hat{k})$ is equal to

26 Jun 2022 (E)

(1) 6

(2) 7

(3) 8

(4) 9

Q105* If $\vec{a} \cdot \vec{b} = 1$, $\vec{b} \cdot \vec{c} = 2$ and $\vec{c} \cdot \vec{a} = 3$, then the value of $\left[\vec{a} \times (\vec{b} \times \vec{c}) - \vec{b} \times (\vec{c} \times \vec{a}) - \vec{c} \times (\vec{a} \times \vec{b}) \right]$ is

26 Jun 2022 (M)

(1) 0

(2) $-6\vec{a} \cdot (\vec{b} \times \vec{c})$

(3) $12\vec{c} \cdot (\vec{a} \times \vec{b})$

(4) $-12\vec{b} \cdot (\vec{c} \times \vec{a})$

Q106. Let θ be the angle between the vectors $\vec{a}$ and $\vec{b}$, where $|\vec{a}| = 4$, $|\vec{b}| = 3$ and $\theta \in (\frac{\pi}{4}, \frac{\pi}{3})$. Then

25 Jun 2022 (M)

$\left| (\vec{a} - \vec{b}) \times (\vec{a} + \vec{b}) \right|^2 + 4(\vec{a} \cdot \vec{b})^2$ is equal to _____

Q107. Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $a_i > 0$, $i = 1, 2, 3$ be a vector which makes equal angles with the coordinate axes

OX , OY and OZ . Also, let the projection of $\vec{a}$ on the vector $3\hat{i} + 4\hat{j}$ be 7. Let $\vec{b}$ be a vector obtained by rotating $\vec{a}$ with 90° . If $\vec{a}$, $\vec{b}$ and x -axis are coplanar, then projection of $\vec{a}$ vector $\vec{b}$ on $3\hat{i} + 4\hat{j}$ is equal to

25 Jun 2022 (M)

(1) $\sqrt{7}$

(2) $\sqrt{2}$

(3) 2

(4) 7

Q108. Let $\vec{b} = \hat{i} + \hat{j} + \lambda\hat{k}$, $\lambda \in \mathbb{R}$. If $\vec{a}$ is a vector such that $\vec{a} \times \vec{b} = 13\hat{i} - \hat{j} - 4\hat{k}$ and $\vec{a} \cdot \vec{b} + 21 = 0$, then

25 Jun 2022 (E)

$(\vec{b} - \vec{a}) \cdot (\hat{k} - \hat{j}) + (\vec{b} + \vec{a}) \cdot (\hat{i} - \hat{k})$ is equal to

Q109. Let $\hat{a}$, $\hat{b}$ be unit vectors. If $\vec{c}$ be a vector such that the angle between $\hat{a}$ and $\vec{c}$ is $\frac{\pi}{12}$, and $\hat{b} = \vec{c} + 2(\vec{c} \times \hat{a})$,

24 Jun 2022 (M)

then $|\vec{c}|^2$ is equal to:

(1) $6(3 - \sqrt{3})$

(2) $6(3 + \sqrt{3})$

(3) $3 + \sqrt{3}$

(4) $6(\sqrt{3} + 1)$

Q110. Let a and b be two unit vectors such that $|(\hat{a} + \hat{b}) + 2(\hat{a} \times \hat{b})| = 2$. If $\theta \in (0, \pi)$ is the angle between $\hat{a}$ and $\hat{b}$, then among the statement

24 Jun 2022 (E)

(S1) : $2\hat{a} \times \hat{b} = \hat{a} - \hat{b}$

(S2) : The projection of $\hat{a}$ on $(\hat{a} + \hat{b})$ is $\frac{1}{2}$

(1) Only (S1) is true.

(2) Only (S2) is true.

(3) Both (S1) and (S2) are true.

(4) Both (S1) and (S2) are false.

Q111. Let $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$. Let a vector $\vec{v}$ be in the plane containing $\vec{a}$ and $\vec{b}$. If $\vec{v}$ is perpendicular to the vector $3\hat{i} + 2\hat{j} - \hat{k}$ and its projection on $\vec{a}$ is 19 units, then $|\vec{v}|^2$ is equal to _____.

01 Sep 2021 (E)

Q112. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$ and the angle between $\vec{a}$ and $\vec{b}$ is 60° . If $\frac{1}{8}\vec{a}$ is a unit vector, then $|\vec{b}|$ is equal to :

31 Aug 2021 (M)

(1) 8

(2) 4

(3) 6

(4) 5

Q113* Let $\vec{a}$, $\vec{b}$, $\vec{c}$ be three vectors mutually perpendicular to each other and have same magnitude. If a vector $\vec{r}$ satisfies $\vec{a} \times \left\{ (\vec{r} - \vec{b}) \times \vec{a} \right\} + \vec{b} \times \left\{ (\vec{r} - \vec{c}) \times \vec{b} \right\} + \vec{c} \times \left\{ (\vec{r} - \vec{a}) \times \vec{c} \right\} = \vec{0}$, then $\vec{r}$ is equal to:

31 Aug 2021 (E)

(1) $\frac{1}{3}(\vec{a} + \vec{b} + \vec{c})$

(2) $\frac{1}{3}(2\vec{a} + \vec{b} - \vec{c})$

(3) $\frac{1}{2}(\vec{a} + \vec{b} + \vec{c})$

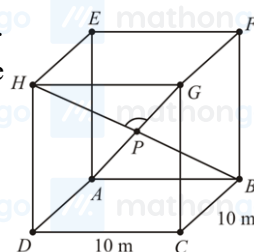
(4) $\frac{1}{2}(\vec{a} + \vec{b} + 2\vec{c})$

Q114. Let $\vec{a} = \hat{i} + 5\hat{j} + \alpha\hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} + \beta\hat{k}$ and $\vec{c} = -\hat{i} + 2\hat{j} - 3\hat{k}$ be three vectors such that, $|\vec{b} \times \vec{c}| = 5\sqrt{3}$ and $\vec{a}$ is perpendicular to $\vec{b}$. Then the greatest amongst the values of $|\vec{a}|^2$ is _____.

27 Aug 2021 (M)

Q115. A hall has a square floor of dimension 10 m $\times$ 10 m (see the figure) and vertical walls.

If the angle GPH between the diagonals AG and BH is $\cos^{-1} \frac{1}{5}$, then the height of the hall (in meters) is:



26 Aug 2021 (E)

(1) $5\sqrt{2}$

(2) $5\sqrt{3}$

(3) $5\sqrt{10}$

(4) 5

Q116. If the projection of the vector $\hat{i} + 2\hat{j} + \hat{k}$ on the sum of the two vectors $2\hat{i} + 4\hat{j} - 5\hat{k}$ and $-\lambda\hat{i} + 2\hat{j} + 3\hat{k}$ is 1, then λ is equal to _____.

26 Aug 2021 (E)

Q117* Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{j} - \hat{k}$. If $\vec{c}$ is a vector such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$, then $\vec{a} \cdot (\vec{b} \times \vec{c})$ is equal to:

26 Aug 2021 (M)

(1) 6

(2) -2

(3) 2

(4) -6

Q118. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{j} - \hat{k}$ be three vectors such that $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{a} \cdot \vec{b} = 1$. If the length of projection vector of the vector $\vec{b}$ on the vector $\vec{a} \times \vec{c}$ is l , then the value of $3l^2$ is equal to _____.

27 Jul 2021 (M)

Q119* Let $\vec{a} = \hat{i} - \alpha\hat{j} + \beta\hat{k}$, $\vec{b} = 3\hat{i} + \beta\hat{j} - \alpha\hat{k}$ and $\vec{c} = -\alpha\hat{i} - 2\hat{j} + \hat{k}$, where α and β are integers. If $\vec{a} \cdot \vec{b} = -1$ and $\vec{b} \cdot \vec{c} = 10$, then $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is equal to _____.

27 Jul 2021 (E)

Q120* Let $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} + 2\hat{j} + 3\hat{k}$. Then the vector product $\left(\frac{\vec{a} + \vec{b}}{2}\right) \times \left(\left(\frac{\vec{a}}{2} \times \left(\left(\frac{\vec{a} - \vec{b}}{2}\right) \times \vec{b}\right)\right) \times \vec{b}\right)$ is equal to :

27 Jul 2021 (M)

- (1) $5(34\hat{i} - 5\hat{j} + 3\hat{k})$ (2) $7(34\hat{i} - 5\hat{j} + 3\hat{k})$
(3) $7(30\hat{i} - 5\hat{j} + 7\hat{k})$ (4) $5(30\hat{i} - 5\hat{j} + 7\hat{k})$

Q121* Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors such that $\vec{a} = \vec{b} \times (\vec{b} \times \vec{c})$. If magnitudes of the vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are $\sqrt{2}, 1$ and 2 respectively and the angle between $\vec{b}$ and $\vec{c}$ is $\theta (0 < \theta < \frac{\pi}{2})$, then the value of $1 + \tan \theta$ is equal to :

27 Jul 2021 (E)

- (1) $\sqrt{3} + 1$ (2) 2
(3) 1 (4) $\frac{\sqrt{3}+1}{\sqrt{3}}$

Q122. If $|\vec{a}| = 2, |\vec{b}| = 5$ and $|\vec{a} \times \vec{b}| = 8$, then $|\vec{a} \cdot \vec{b}|$ is equal to:

25 Jul 2021 (E)

- (1) 6 (2) 4
(3) 3 (4) 5

Q123. Let the vectors $(2 + a + b)\hat{i} + (a + 2b + c)\hat{j} - (b + c)\hat{k}, (1 + b)\hat{i} + 2b\hat{j} - b\hat{k}$ and $(2 + b)\hat{i} + 2b\hat{j} + (1 - b)\hat{k}, \forall a, b, c \in R$ be co-planar. Then which of the following is true?

25 Jul 2021 (M)

- (1) $2b = a + c$ (2) $3c = a + b$
(3) $a = b + 2c$ (4) $2a = b + c$

Q124. Let a, b and c be distinct positive numbers. If the vectors $a\hat{i} + a\hat{j} + c\hat{k}, \hat{i} + \hat{k}$ and $c\hat{i} + c\hat{j} + b\hat{k}$ are co-planar, then c is equal to:

25 Jul 2021 (E)

- (1) $\frac{2}{\frac{1}{a} + \frac{1}{b}}$ (2) $\frac{a+b}{2}$
(3) $\frac{1}{a} + \frac{1}{b}$ (4) $\sqrt{ab}$

Q125. Let $\vec{p} = 2\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{q} = \hat{i} + 2\hat{j} + \hat{k}$ be two vectors. If a vector $\vec{r} = (\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k})$ is perpendicular to each of the vectors $(\vec{p} + \vec{q})$ and $(\vec{p} - \vec{q})$, and $|\vec{r}| = \sqrt{3}$, then $|\alpha| + |\beta| + |\gamma|$ is equal to _____

25 Jul 2021 (M)

Q126. If $\vec{a}$ and $\vec{b}$ are unit vectors and $(\vec{a} + 3\vec{b})$ is perpendicular to $(7\vec{a} - 5\vec{b})$ and $(\vec{a} - 4\vec{b})$ is perpendicular to $(7\vec{a} - 2\vec{b})$, then the angle between $\vec{a}$ and $\vec{b}$ (in degrees) is _____.

25 Jul 2021 (E)

Q127* Let a vector $\vec{a}$ be coplanar with vectors $\vec{b} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} + \hat{k}$. If $\vec{a}$ is perpendicular to $\vec{d} = 3\hat{i} + 2\hat{j} + 6\hat{k}$, and $|\vec{a}| = \sqrt{10}$. Then a possible value of $\left[\frac{\vec{a}}{|\vec{a}|} \frac{\vec{b}}{|\vec{b}|} \frac{\vec{c}}{|\vec{c}|}\right] + \left[\frac{\vec{a}}{|\vec{a}|} \frac{\vec{b}}{|\vec{b}|} \frac{\vec{d}}{|\vec{d}|}\right] + \left[\frac{\vec{a}}{|\vec{a}|} \frac{\vec{c}}{|\vec{c}|} \frac{\vec{d}}{|\vec{d}|}\right]$ is equal to:

22 Jul 2021 (M)

- (1) -42 (2) -40
(3) -29 (4) -38

Q128* Let three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ be such that $\vec{a} \times \vec{b} = \vec{c}, \vec{b} \times \vec{c} = \vec{a}$ and $|\vec{a}| = 2$. Then which one of the following is not true? 22 Jul 2021 (M)

- (1) $\vec{a} \times \left(\left(\vec{b} + \vec{c} \right) \times \left(\vec{b} - \vec{c} \right) \right) = \vec{0}$ (2) Projection of $\vec{a}$ on $(\vec{b} \times \vec{c})$ is 2
(3) $\left[\vec{a} \vec{b} \vec{c} \right] + \left[\vec{c} \vec{a} \vec{b} \right] = 8$ (4) $\left| 3\vec{a} + \vec{b} - 2\vec{c} \right|^2 = 51$

Q129. Let $\vec{a}, \vec{b}, \vec{c}$ be three mutually perpendicular vectors of the same magnitude and equally inclined at an angle θ , with the vector $\vec{a} + \vec{b} + \vec{c}$. Then $36 \cos^2 2\theta$ is equal to 20 Jul 2021 (M)

Q130. In a triangle ABC , if $|\vec{BC}| = 3, |\vec{CA}| = 5$ and $|\vec{BA}| = 7$, then the projection of the vector $\vec{BA}$ on $\vec{BC}$ is equal to 20 Jul 2021 (E)

(1) $\frac{19}{2}$ (2) $\frac{13}{2}$
(3) $\frac{11}{2}$ (4) $\frac{15}{2}$

Q131. Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = |\vec{c}|, |\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $(\vec{a} \times \vec{b})$ and $\vec{c}$ is $\frac{\pi}{6}$, then the value of $\left| (\vec{a} \times \vec{b}) \times \vec{c} \right|$ is: 20 Jul 2021 (M)

(1) $\frac{2}{3}$ (2) 4
(3) 3 (4) $\frac{3}{2}$

Q132. For $p > 0$, a vector $\vec{v}_2 = 2\hat{i} + (p+1)\hat{j}$ is obtained by rotating the vector $\vec{v}_1 = \sqrt{3}p\hat{i} + \hat{j}$ by an angle θ about origin in counter clockwise direction. If $\tan \theta = \frac{(\alpha\sqrt{3}-2)}{(4\sqrt{3}+3)}$, then the value of α is equal to 20 Jul 2021 (E)

Q133. Let $\vec{a}$ and $\vec{b}$ be two non-zero vectors perpendicular to each other and $|\vec{a}| = |\vec{b}|$. If $|\vec{a} \times \vec{b}| = |\vec{a}|$, then the angle between the vectors $\left(\vec{a} + \vec{b} + \left(\vec{a} \times \vec{b} \right) \right)$ and $\vec{a}$ is equal to : 18 Mar 2021 (E)

(1) $\sin^{-1} \left(\frac{1}{\sqrt{3}} \right)$ (2) $\cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$
(3) $\cos^{-1} \left(\frac{1}{\sqrt{2}} \right)$ (4) $\sin^{-1} \left(\frac{1}{\sqrt{6}} \right)$

Q134. In a triangle ABC , if $|\vec{BC}| = 8, |\vec{CA}| = 7, |\vec{AB}| = 10$, then the projection of the vector $\vec{AB}$ on $\vec{AC}$ is equal to : 18 Mar 2021 (E)

(1) $\frac{25}{4}$ (2) $\frac{85}{14}$
(3) $\frac{127}{20}$ (4) $\frac{115}{16}$

Q135. A vector $\vec{a}$ has components $3p$ and 1 with respect to a rectangular cartesian system. This system is rotated through a certain angle about the origin in the counter clockwise sense. If, with respect to new system, $\vec{a}$ has components $p+1$ and $\sqrt{10}$, then a value of p is equal to: 18 Mar 2021 (M)

(1) 1 (2) $-\frac{5}{4}$
(3) $\frac{4}{5}$ (4) -1

Q136. Let $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{b} = 7\hat{i} + \hat{j} - 6\hat{k}$. If $\vec{r} \times \vec{a} = \vec{r} \times \vec{b}$, $\vec{r} \cdot (\hat{i} + 2\hat{j} + \hat{k}) = -3$, then $\vec{r} \cdot (2\hat{i} - 3\hat{j} + \hat{k})$ is equal to: 17 Mar 2021 (M)

(1) 12

(2) 8

(3) 13

(4) 10

Q137. Let $\vec{x}$ be a vector in the plane containing vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$. If the vector $\vec{x}$ is perpendicular to $(3\hat{i} + 2\hat{j} - \hat{k})$ and its projection on $\vec{a}$ is $\frac{17\sqrt{6}}{2}$, then the value of $|\vec{x}|$ is equal to _____. 17 Mar 2021 (E)

Q138. Let O be the origin. Let $\vec{OP} = x\hat{i} + y\hat{j} - \hat{k}$ and $\vec{OQ} = -\hat{i} + 2\hat{j} + 3x\hat{k}$, $x, y \in R$, $x > 0$, be such that $|\vec{PQ}| = \sqrt{20}$ and the vector $\vec{OP}$ is perpendicular to $\vec{OQ}$. If $\vec{OR} = 3\hat{i} + z\hat{j} - 7\hat{k}$, $z \in R$, is coplanar with $\vec{OP}$ and $\vec{OQ}$, then the value of $x^2 + y^2 + z^2$ is equal to 17 Mar 2021 (E)

(1) 7

(2) 9

(3) 2

(4) 1

Q139*. If $\vec{a} = \alpha\hat{i} + \beta\hat{j} + 3\hat{k}$, $\vec{b} = -\beta\hat{i} - \alpha\hat{j} - \hat{k}$ and $\vec{c} = \hat{i} - 2\hat{j} - \hat{k}$ such that $\vec{a} \cdot \vec{b} = 1$ and $\vec{b} \cdot \vec{c} = -3$, then $\frac{1}{3} \left((\vec{a} \times \vec{b}) \cdot \vec{c} \right)$ is equal to _____. 17 Mar 2021 (M)

Q140. Let a vector $\alpha\hat{i} + \beta\hat{j}$ be obtained by rotating the vector $\sqrt{3}\hat{i} + \hat{j}$ by an angle 45° about the origin in counterclockwise direction in the first quadrant. Then the area (in sq. units) of triangle having vertices (α, β) , $(0, \beta)$ and $(0, 0)$ is equal to 16 Mar 2021 (M)

(1) $\frac{1}{2}$

(2) 1

(3) $\frac{1}{\sqrt{2}}$ (4) $2\sqrt{2}$

Q141. Let $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 2\hat{i} - 3\hat{j} + 5\hat{k}$. If $\vec{r} \times \vec{a} = \vec{b} \times \vec{r}$, $\vec{r} \cdot (\alpha\hat{i} + 2\hat{j} + \hat{k}) = 3$ and $\vec{r} \cdot (2\hat{i} + 5\hat{j} - \alpha\hat{k}) = -1$, $\alpha \in R$, then the value of $\alpha + |\vec{r}|^2$ is equal to : 16 Mar 2021 (E)

(1) 9

(2) 15

(3) 13

(4) 11

Q142. Let $\vec{c}$ be a vector perpendicular to the vectors $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. If $\vec{c} \cdot (\hat{i} + \hat{j} + 3\hat{k}) = 8$, then the value of $\vec{c} \cdot (\vec{a} \times \vec{b})$ is equal to 16 Mar 2021 (E)

Q143. If vectors $\vec{a}_1 = x\hat{i} - \hat{j} + \hat{k}$ and $\vec{a}_2 = \hat{i} + y\hat{j} + z\hat{k}$ are collinear, then a possible unit vector parallel to the vector $x\hat{i} + y\hat{j} + z\hat{k}$ is: 26 Feb 2021 (E)

(1) $\frac{1}{\sqrt{2}}(-\hat{j} + \hat{k})$ (2) $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} - \hat{k})$ (3) $\frac{1}{\sqrt{2}}(\hat{i} - \hat{j})$ (4) $\frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$

Q144*. If $\vec{a}$ and $\vec{b}$ are perpendicular, then $\vec{a} \times \left(\vec{a} \times \left(\vec{a} \times \left(\vec{a} \times \vec{b} \right) \right) \right)$ is equal to: 26 Feb 2021 (M)

(1) $|\vec{a}|^4 \vec{b}$

(2) $\vec{0}$

(3) $\vec{a} \times \vec{b}$

(4) $\frac{1}{2} |\vec{a}|^4 \vec{b}$

Q145. If $(1, 5, 35)$, $(7, 5, 5)$, $(1, \lambda, 7)$ and $(2\lambda, 1, 2)$ are coplanar, then the sum of all possible values of λ is:

26 Feb 2021 (M)

(1) $\frac{44}{5}$

(2) $-\frac{44}{5}$

(3) $\frac{39}{5}$

(4) $-\frac{39}{5}$

Q146. Let $\vec{a} = \hat{i} + \alpha\hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} - \alpha\hat{j} + \hat{k}$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $8\sqrt{3}$ square units, then $\vec{a} \cdot \vec{b}$ is equal to ____.

25 Feb 2021 (E)

Q147. Let $\vec{a} = \hat{i} + 2\hat{j} - \hat{k}$, $\vec{b} = \hat{i} - \hat{j}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = \vec{c} \times \vec{a}$ and $\vec{r} \cdot \vec{b} = 0$, then $\vec{r} \cdot \vec{a}$ is equal to

25 Feb 2021 (M)

Q148*. Let three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ be such that $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$, $\vec{a} \cdot \vec{c} = 7$ and $\vec{b}$ is perpendicular to $\vec{c}$, where $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{k}$, then the value of $2|\vec{a} + \vec{b} + \vec{c}|^2$ is

24 Feb 2021 (M)

Q149. If $\vec{x}$ and $\vec{y}$ be two non-zero vectors such that $|\vec{x} + \vec{y}| = |\vec{x}|$ and $2\vec{x} + \lambda\vec{y}$ is perpendicular to $\vec{y}$, then the value of λ is

06 Sep 2020 (E)

Q150. If $\vec{a}$ and $\vec{b}$ are unit vectors, then the greatest value of $\sqrt{3}|\vec{a} + \vec{b}| + |\vec{a} - \vec{b}|$ is

06 Sep 2020 (M)

Q151. Let the vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ be such that $|\vec{a}| = 2$, $|\vec{b}| = 4$ and $|\vec{c}| = 4$. If the projection of $\vec{b}$ on $\vec{a}$ is equal to the projection of $\vec{c}$ on $\vec{a}$ and $\vec{b}$ is perpendicular to $\vec{c}$, then the value of $|\vec{a} + \vec{b} - \vec{c}|$ is ...

05 Sep 2020 (E)

Q152*. If the volume of a parallelepiped, whose coterminous edges are given by the vectors $\vec{a} = \hat{i} + \hat{j} + n\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} - n\hat{k}$ and $\vec{c} = \hat{i} + n\hat{j} + 3\hat{k}$ ($n \geq 0$) is 158 cubic units, then :

05 Sep 2020 (M)

(1) $\vec{a} \cdot \vec{c} = 17$

(2) $\vec{b} \cdot \vec{c} = 10$

(3) $n = 7$

(4) $n = 9$

Q153*. If $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$, then the value of $|\hat{i} \times (\vec{a} \times \hat{i})|^2 + |\hat{j} \times (\vec{a} \times \hat{j})|^2 + |\hat{k} \times (\vec{a} \times \hat{k})|^2$, is equal to :

04 Sep 2020 (E)

Q154*. Let x_0 be the point of local maxima of $f(x) = \vec{a} \cdot \left(\vec{b} \times \vec{c} \right)$, where $\vec{a} = x\hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = -2\hat{i} + x\hat{j} - \hat{k}$ and $\vec{c} = 7\hat{i} - 2\hat{j} + x\hat{k}$. Then the value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ at $x = x_0$ is:

04 Sep 2020 (M)

(1) -4

(3) 14

(2) -30

(4) -22

Q155. Let $a, b, c \in R$ be such that $a^2 + b^2 + c^2 = 1$. If $a \cos \theta = b \cos(\theta + \frac{2\pi}{3}) = c \cos(\theta + \frac{4\pi}{3})$, where $\theta = \frac{\pi}{9}$, then the angle between the vectors $a\hat{i} + b\hat{j} + c\hat{k}$ and $b\hat{i} + c\hat{j} + a\hat{k}$ is:

03 Sep 2020 (E)

(1) 0

(3) $\frac{\pi}{2}$

(2) $\frac{2\pi}{3}$

(4) $\frac{\pi}{9}$

Q156. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three unit vectors such that $|\vec{a} - \vec{b}|^2 + |\vec{a} - \vec{c}|^2 = 8$. Then $|\vec{a} + 2\vec{b}|^2 + |\vec{a} + 2\vec{c}|^2$ is equal to

02 Sep 2020 (M)

Q157. Let the position vectors of points 'A' and 'B' be $\hat{i} + \hat{j} + \hat{k}$ and $2\hat{i} + \hat{j} + 3\hat{k}$, respectively. A point 'P' divides the line segment AB internally in the ratio $\lambda : 1 (\lambda > 0)$. If O is the origin and $\vec{OB} \cdot \vec{OP} - 3|\vec{OA} \times \vec{OP}|^2 = 6$

then λ is equal to

02 Sep 2020 (E)

Q158. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors such that $|\vec{a}| = \sqrt{3}, |\vec{b}| = 5, \vec{b} \cdot \vec{c} = 10$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$. If $\vec{a}$ is perpendicular to the vector $\vec{b} \times \vec{c}$, then $|\vec{a} \times (\vec{b} \times \vec{c})|$ is equal to _____.

09 Jan 2020 (E)

Q159. If the vectors, $\vec{p} = (a+1)\hat{i} + a\hat{j} + a\hat{k}, \vec{q} = a\hat{i} + (a+1)\hat{j} + a\hat{k}$ and $\vec{r} = a\hat{i} + a\hat{j} + (a+1)\hat{k} (a \in R)$ are coplanar and $3(\vec{p} \cdot \vec{q})^2 - \lambda |\vec{r} \times \vec{q}|^2 = 0$, then the value of λ is _____.

09 Jan 2020 (M)

Q160. Let $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, be two vectors. If $\vec{c}$, is a vector such that $\vec{b} \times \vec{c} = \vec{b} \times \vec{a}$ and $\vec{c} \cdot \vec{a} = 0$, then $\vec{c} \cdot \vec{b}$, is equal to.

08 Jan 2020 (E)

(1) $-\frac{3}{2}$

(3) $-\frac{1}{2}$

(2) $\frac{1}{2}$

(4) -1

Q161*. Let the volume of a parallelepiped whose coterminal edges are given by $\vec{u} = \hat{i} + \hat{j} + \lambda\hat{k}, \vec{v} = \hat{i} + \hat{j} + 3\hat{k}$ and $\vec{w} = 2\hat{i} + \hat{j} + \hat{k}$ be 1 cu. unit. If θ be the angle between the edges $\vec{u}$ and $\vec{w}$, then the value of $\cos \theta$ can be

08 Jan 2020 (M)

(1) $\frac{7}{6\sqrt{6}}$

(3) $\frac{5}{7}$

(2) $\frac{7}{6\sqrt{3}}$

(4) $\frac{5}{3\sqrt{3}}$

Q162. A vector $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k} (\alpha, \beta \in R)$ lies in the plane of the vectors, $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{i} - \hat{j} + 4\hat{k}$. If $\vec{a}$ bisects the angle between $\vec{b}$ and $\vec{c}$, then

07 Jan 2020 (M)

(1) $\vec{a} \cdot \hat{i} + 3 = 0$

(3) $\vec{a} \cdot \hat{k} + 2 = 0$

(2) $\vec{a} \cdot \hat{i} + 1 = 0$

(4) $\vec{a} \cdot \hat{k} + 4 = 0$

Q163. Let $\vec{a}, \vec{b}$ and $\vec{c}$, be three unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. If $\lambda = \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ and $\vec{d} = \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$, then the order pair, $(\lambda, \vec{d})$, is equal to. 07 Jan 2020 (E)

- (1) $\left(\frac{3}{2}, 3\vec{a} \times \vec{c}\right)$ (2) $\left(-\frac{3}{2}, 3\vec{c} \times \vec{b}\right)$
 (3) $\left(\frac{3}{2}, 3\vec{b} \times \vec{c}\right)$ (4) $\left(-\frac{3}{2}, 3\vec{a} \times \vec{b}\right)$

Q164. Let $\alpha \in R$ and the three vectors $\vec{a} = \alpha\hat{i} + \hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} + \hat{j} - \alpha\hat{k}$ and $\vec{c} = \alpha\hat{i} - 2\hat{j} + 3\hat{k}$. Then the set $S = \{\alpha : \vec{a}, \vec{b} \text{ and } \vec{c} \text{ are coplanar}\}$ 12 Apr 2019 (E)

- (1) is singleton
 (2) contains exactly two positive numbers
 (3) is empty
 (4) contains exactly two numbers only one of which is positive

Q165. Let $\vec{a} = 3\hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$ be two vectors. If a vector perpendicular to both the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ has the magnitude 12 then one such vector is: 12 Apr 2019 (M)

- (1) $4(2\hat{i} + 2\hat{j} + \hat{k})$ (2) $4(2\hat{i} - 2\hat{j} - \hat{k})$
 (3) $4(-2\hat{i} - 2\hat{j} + \hat{k})$ (4) $4(2\hat{i} + 2\hat{j} - \hat{k})$

Q166*. If the volume of parallelepiped formed by the vectors $\hat{i} + \lambda\hat{j} + \hat{k}$, $\hat{j} + \lambda\hat{k}$ and $\lambda\hat{i} + \hat{k}$ is minimum, then λ is equal to: 12 Apr 2019 (M)

- (1) $-\frac{1}{\sqrt{3}}$ (2) $-\sqrt{3}$
 (3) $\sqrt{3}$ (4) $\frac{1}{\sqrt{3}}$

Q167. Let $A(3, 0, -1)$, $B(2, 10, 6)$ and $C(1, 2, 1)$ be the vertices of a triangle and M be the mid-point of AC . If G divides BM in the ratio, $2 : 1$, then $\cos(\angle GOA)$ (O being the origin) is equal to 10 Apr 2019 (M)

- (1) $\frac{1}{\sqrt{30}}$ (2) $\frac{1}{6\sqrt{10}}$
 (3) $\frac{1}{\sqrt{15}}$ (4) $\frac{1}{2\sqrt{15}}$

Q168. The distance of the point having position vector $-\hat{i} + 2\hat{j} + 6\hat{k}$ from the straight line passing through the point $(2, 3, -4)$ and parallel to the vector, $6\hat{i} + 3\hat{j} - 4\hat{k}$ is 10 Apr 2019 (E)

- (1) $4\sqrt{3}$ (2) 6
 (3) $2\sqrt{13}$ (4) 7

Q169. If a unit vector $\vec{a}$ makes angles $\frac{\pi}{3}$ with $\hat{i}$, $\frac{\pi}{4}$ with $\hat{j}$ and $\theta \in (0, \pi)$ with $\hat{k}$, then a value of θ is: 09 Apr 2019 (E)

- (1) $\frac{5\pi}{6}$ (2) $\frac{5\pi}{12}$
 (3) $\frac{\pi}{4}$ (4) $\frac{2\pi}{3}$

Q170. Let $\vec{\alpha} = 3\hat{i} + \hat{j}$ and $\vec{\beta} = 2\hat{i} - \hat{j} + 3\hat{k}$. If $\vec{\beta} = \vec{\beta}_1 - \vec{\beta}_2$, where $\vec{\beta}_1$ is parallel to $\vec{\alpha}$ and $\vec{\beta}_2$ is perpendicular to $\vec{\alpha}$, then $\vec{\beta}_1 \times \vec{\beta}_2$ is equal to: 09 Apr 2019 (M)

(1) $\frac{1}{2}(-3\hat{i} + 9\hat{j} + 5\hat{k})$

(3) $-3\hat{i} + 9\hat{j} + 5\hat{k}$

(2) $3\hat{i} - 9\hat{j} - 5\hat{k}$

(4) $\frac{1}{2}(3\hat{i} - 9\hat{j} + 5\hat{k})$

Q171. Let $\vec{a} = 3\hat{i} + 2\hat{j} + x\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, for some real x . Then the condition for $|\vec{a} \times \vec{b}| = r$ to follow

08 Apr 2019 (E)

(1) $0 < r \leq \sqrt{\frac{3}{2}}$

(2) $r \geq 5\sqrt{\frac{3}{2}}$

(3) $\sqrt{\frac{3}{2}} < r \leq 3\sqrt{\frac{3}{2}}$

(4) $3\sqrt{\frac{3}{2}} < r < 5\sqrt{\frac{3}{2}}$

Q172. The magnitude of the projection of the vector $2\hat{i} + 3\hat{j} + \hat{k}$ on the vector perpendicular to the plane containing the vectors $\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} + 3\hat{k}$, is:

08 Apr 2019 (M)

(1) $3\sqrt{6}$

(2) $\sqrt{\frac{3}{2}}$

(3) $\sqrt{6}$

(4) $\frac{\sqrt{3}}{2}$

Q173. The sum of the distinct real values of μ for which the vectors $\mu\hat{i} + \hat{j} + \hat{k}$, $\hat{i} + \mu\hat{j} + \hat{k}$, $\hat{i} + \hat{j} + \mu\hat{k}$ are coplanar, is

12 Jan 2019 (M)

(1) 0

(2) -1

(3) 1

(4) 2

Q174*. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three unit vectors, out of which vectors $\vec{b}$ and $\vec{c}$ are non-parallel. If α and β are the angles which vector $\vec{a}$ makes with vectors $\vec{b}$ and $\vec{c}$ respectively and $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{1}{2}\vec{b}$, then $|\alpha - \beta|$ is equal to :

12 Jan 2019 (E)

(1) 90°

(2) 60°

(3) 45°

(4) 30°

Q175. Let $\vec{a} = \hat{i} + 2\hat{j} + 4\hat{k}$, $\vec{b} = \hat{i} + \lambda\hat{j} + 4\hat{k}$ and $\vec{c} = 2\hat{i} + 4\hat{j} + (\lambda^2 - 1)\hat{k}$ be coplanar vectors. Then the non-zero vector $\vec{a} \times \vec{c}$ is:

11 Jan 2019 (M)

(1) $-10\hat{i} - 5\hat{j}$

(2) $-14\hat{i} - 5\hat{j}$

(3) $-14\hat{i} + 5\hat{j}$

(4) $-10\hat{i} + 5\hat{j}$

Q176. Let $\sqrt{3}\hat{i} + \hat{j}$, $\hat{i} + \sqrt{3}\hat{j}$ and $\beta\hat{i} + (1 - \beta)\hat{j}$ respectively be the position vectors of the points A, B and C with respect to the origin O. If the distance of C from the bisector of the acute angle between OA and OB is $\frac{3}{\sqrt{2}}$, then the sum of all possible values of β is:

11 Jan 2019 (E)

(1) 4

(2) 3

(3) 2

(4) 1

Q177. Let $\vec{\alpha} = (\lambda - 2)\vec{a} + \vec{b}$ and $\vec{\beta} = (4\lambda - 2)\vec{a} + 3\vec{b}$, be two given vectors where vectors $\vec{a}$ and $\vec{b}$ are non-collinear. The value of λ for which vectors $\vec{\alpha}$ and $\vec{\beta}$ are collinear, is:

10 Jan 2019 (E)

(1) -4

(2) -3

(3) 4

(4) 3

Q178. Let $\vec{a} = 2\hat{i} + \lambda_1\hat{j} + 3\hat{k}$, $\vec{b} = 4\hat{i} + (3 - \lambda_2)\hat{j} + 6\hat{k}$ and $\vec{c} = 3\hat{i} + 6\hat{j} + (\lambda_3 - 1)\hat{k}$ be three vectors such that $\vec{b} = 2\vec{a}$ and $\vec{a}$ is perpendicular to $\vec{c}$. Then a possible value of $(\lambda_1, \lambda_2, \lambda_3)$ is **10 Jan 2019 (M)**

(1) $(-\frac{1}{2}, 4, 0)$

(2) $(1, 5, 1)$

(3) $(\frac{1}{2}, 4, -2)$

(4) $(1, 3, 1)$

Q179. Let A be a point on the line $\vec{r} = (1 - 3\mu)\hat{i} + (\mu - 1)\hat{j} + (2 + 5\mu)\hat{k}$ and $B(3, 2, 6)$ be a point in the space.

Then the value of μ for which the vector $\vec{AB}$ is parallel to the plane $x - 4y + 3z = 1$ is **10 Jan 2019 (M)**

(1) $\frac{1}{2}$

(2) $\frac{1}{4}$

(3) $-\frac{1}{4}$

(4) $\frac{1}{8}$

Q180. Let $\vec{a} = \hat{i} + \hat{j} + \sqrt{2}\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + \sqrt{2}\hat{k}$ and $\vec{c} = 5\hat{i} + \hat{j} + \sqrt{2}\hat{k}$ be three vectors such that the projection vector of $\vec{b}$ on $\vec{a}$ is $\left|\vec{a}\right|$. If $\vec{a} + \vec{b}$ is perpendicular to $\vec{c}$, then $\left|\vec{b}\right|$ is equal to: **09 Jan 2019 (E)**

(1) $\sqrt{22}$

(2) $\sqrt{32}$

(3) 6

(4) 4

Q181*. Let $\vec{a} = \hat{i} - \hat{j}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} \times \vec{c} + \vec{b} = \vec{0}$ and $\vec{a} \cdot \vec{c} = 4$, then $\left|\vec{c}\right|^2$ is equal to:

(1) $\frac{19}{2}$

(2) 9

(3) $\frac{17}{2}$

(4) 8

ANSWER KEYS

1. (2)	2. (4)	3. (1)	4. (1)	5. (3)	6. (3)	7. (2)	8. (569)
9. (46)	10. (3)	11. (3)	12. (2)	13. (30)	14. (4)	15. (1)	16. (54)
17. (3)	18. (1)	19. (1)	20. (1)	21. (38)	22. (4)	23. (48)	24. (38)
25. (2)	26. (3)	27. (1)	28. (2)	29. (1)	30. (4)	31. (2)	32. (2)
33. (5)	34. (4)	35. (3)	36. (4)	37. (1)	38. (66)	39. (4)	40. (4)
41. (1)	42. (288)	43. (285)	44. (4)	45. (3)	46. (3)	47. (2)	48. (3)
49. (1)	50. (2)	51. (1)	52. (2)	53. (3)	54. (11)	55. (2)	56. (4)
57. (3)	58. (3)	59. (1)	60. (11)	61. (1)	62. (3501)	63. (36)	64. (3)
65. (2)	66. (3)	67. (2)	68. (3)	69. (1)	70. (3)	71. (2)	72. (1)
73. (1)	74. (1)	75. (1)	76. (2)	77. (4)	78. (3)	79. (2)	80. (8)
81. (1)	82. (4)	83. (14)	84. (1)	85. (3)	86. (1)	87. (3)	88. (2)
89. (2)	90. (4)	91. (2)	92. (36)	93. (4)	94. (3)	95. (1)	96. (1)
97. (29)	98. (1)	99. (3)	100. (150)	101. (4)	102. (4)	103. (1)	104. (4)
105. (1)	106. (576)	107. (2)	108. (14)	109. (2)	110. (3)	111. (1494)	112. (4)
113. (3)	114. (90)	115. (1)	116. (5)	117. (2)	118. (2)	119. (9)	120. (2)
121. (2)	122. (1)	123. (1)	124. (4)	125. (3)	126. (60)	127. (1)	128. (4)
129. (4)	130. (3)	131. (4)	132. (6)	133. (2)	134. (2)	135. (4)	136. (1)
137. (486)	138. (2)	139. (2)	140. (1)	141. (2)	142. (28)	143. (4)	144. (1)
145. (1)	146. (2)	147. (12)	148. (75)	149. (1)	150. (4)	151. (6)	152. (2)
153. (18)	154. (4)	155. (3)	156. (2)	157. (0.8)	158. (30)	159. (1)	160. (3)
161. (2)	162. (3)	163. (4)	164. (3)	165. (2)	166. (4)	167. (3)	168. (4)
169. (4)	170. (1)	171. (2)	172. (2)	173. (2)	174. (4)	175. (4)	176. (4)
177. (1)	178. (1)	179. (2)	180. (3)	181. (1)			

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